

Deleting a bracketed pair preserves length-additivity (and the bracket is not needed)

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The commit named above contains the mathematics of this note and the code cited in §5. It depends on earlier work only through [1], whose Lemmas 2.2 and 2.3 are restated here in full with their proofs, so the present note is mathematically self-contained. If this note is edited later, replace this sentence by a statement of what changed by kind — not by a claim that nothing did.

Abstract

Let $S, T \subsetneq \mathbb{Z}/n$ and suppose the factorisation $u_S u_T$ of cyclically decreasing elements of the affine symmetric group is length-additive. Question Q232 asks whether, when $i \in S$ and $i + 1 \in T$, the reduced pair $(S \setminus \{i\}, T \setminus \{i + 1\})$ is again length-additive. We prove that it is. In fact we prove more: the hypothesis $i + 1 \in T$ is never used, so for *every* $i \in S$ the pair $(S \setminus \{i\}, T \setminus \{i + 1\})$ is length-additive, whether or not $i + 1$ lies in T . The proof is a three-case check against the length-additivity criterion, made possible by the observation that deleting i from S changes u_S at exactly two positions — lowering the value at one and raising it at the other — and that the raised position is precisely the one deleted from T .

We then use the theorem to close gap (G1) of the companion note: the $\mathcal{X}(S, T) = \emptyset$ case of the Morse–Schilling two-factor construction, previously recorded as computational, is proved. The transport step turns out to need one further ingredient, a bound $|T| \leq n - 2$ that fails on 894 of the 6786 instances in range; those are exactly the instances where the construction produces no move, and the same three conclusions of the containment theorem that drive the transport also prove it.

1 Conventions, and the two facts we import

Let $n \geq 2$. The affine symmetric group $\tilde{S}_n$ is the group of bijections $w : \mathbb{Z} \rightarrow \mathbb{Z}$ with $w(j + n) = w(j) + n$ and $\sum_{j=1}^n w(j) = \binom{n+1}{2}$; for $i \in \mathbb{Z}/n$, s_i interchanges $i + jn$ and $i + 1 + jn$ for all j . Products act by $(ab)(j) = a(b(j))$ and $\ell(ws_i) = \ell(w) + 1$ iff $w(i) < w(i + 1)$. Every $w \in \tilde{S}_n$ is n -equivariant:

$$w(x + n) = w(x) + n \quad (x \in \mathbb{Z}). \quad (1)$$

A subset $S \subseteq \mathbb{Z}/n$ is identified with its n -periodic lift $\{j \in \mathbb{Z} : j \bmod n \in S\}$, written again S . A *run* of S is a maximal integer interval $[m, M] \subseteq S$; a *gap* is an element of $\mathbb{Z} \setminus S$. If $S \subsetneq \mathbb{Z}/n$ then the gap set is nonempty and n -periodic, so gaps recur with period n and

$$\text{every run of a proper } S \text{ has length } \leq n - 1. \quad (2)$$

The *blocks* of S are the intervals $[g+1, g']$ for consecutive gaps $g < g'$; they partition $\mathbb{Z}$, and a run $[m, M]$ gives the block $[m, M+1]$. For $S \subsetneq \mathbb{Z}/n$ we write u_S for the unique cyclically decreasing element with support S . We call (S, T) *additive* if $S, T \subsetneq \mathbb{Z}/n$ and $\ell(u_S u_T) = |S| + |T|$.

We import two facts from [1, Lems. 2.2, 2.3], restated here with their proofs.

Lemma 1.1 (Block model, [1, Lem. 2.2]). *Let $S \subsetneq \mathbb{Z}/n$. Then u_S rotates each block downwards by one step:*

$$u_S(g+1) = g', \quad u_S(j) = j-1 \quad (g+1 < j \leq g') \quad (3)$$

for every block $[g+1, g']$. Equivalently $u_S(j) = j-1$ if $j-1 \in S$, and $u_S(j) = \min\{c \geq j : c \notin S\}$ otherwise. Moreover $\ell(u_S) = |S|$.

Proof. For a run $[m, M]$ the product $s_M s_{M-1} \cdots s_m$ is a reduced word in distinct generators in which s_{i+1} precedes s_i whenever both occur; as a permutation it sends $m \mapsto M+1$ and $j \mapsto j-1$ for $m < j \leq M+1$, i.e. it is the rotation (3) of the block $[m, M+1]$ and the identity elsewhere. Distinct runs are separated by a gap, so their blocks are disjoint and the rotations commute; the product is the stated permutation, of length $\sum(M-m+1) = |S|$. See [1, Lem. 2.2] for reducedness and uniqueness, which we do not use below. $\square$

Lemma 1.2 (Length-additivity criterion, [1, Lem. 2.3]). *Let $w \in \tilde{S}_n$ and $X \subsetneq \mathbb{Z}/n$. Then $\ell(wu_X) = \ell(w) + |X|$ if and only if for every run $[m, M]$ of X ,*

$$w(M+1) > w(j) \quad \text{for all } j \in [m, M].$$

Proof. For a single run, $u_{[m, M]} = s_M s_{M-1} \cdots s_m$. Multiplying on the right one letter at a time, $\ell(ws_M) = \ell(w) + 1$ iff $w(M) < w(M+1)$; the element ws_M has $M+1$ and M swapped in position, so $\ell(ws_M s_{M-1})$ increases iff $(ws_M)(M-1) < (ws_M)(M)$, i.e. iff $w(M-1) < w(M+1)$; inductively the j -th step increases the length iff $w(M-j+1) < w(M+1)$. Hence the run contributes $M-m+1$ iff $w(M+1) > w(j)$ for all $j \in [m, M]$. Distinct runs act on disjoint position-blocks and commute, and each condition involves only w , so the conditions are simultaneous. By (1) the condition for a run is invariant under replacing $[m, M]$ by $[m+kn, M+kn]$, so it suffices to check one lift of each run. $\square$

2 Deleting one element from S

Fix $S \subsetneq \mathbb{Z}/n$ and $i \in S$; choose an integer lift of i , again written i , and let $[p, q]$ be the run of S containing i , so

$$p \leq i \leq q, \quad p-1 \notin S, \quad q+1 \notin S, \quad q-p \leq n-2 \quad (4)$$

by (2). Write $S' = S \setminus \{i\}$ throughout this section.

Lemma 2.1 (Two-position perturbation). *The maps u_S and $u_{S'}$ agree at every $x \in \mathbb{Z}$ with $x \not\equiv p$ and $x \not\equiv i+1 \pmod{n}$, and*

$$u_S(p) = q+1, \quad u_{S'}(p) = i, \quad u_S(i+1) = i, \quad u_{S'}(i+1) = q+1.$$

In particular $u_{S'}(p) < u_S(p)$ and $u_{S'}(i+1) > u_S(i+1)$.

Proof. The gap set of S' is that of S together with the residue class of i . Hence every block of S other than $[p, q+1]$ and its translates is a block of S' , while $[p, q+1]$ splits into the two blocks $[p, i]$ and $[i+1, q+1]$ (both nonempty, as $p \leq i$ and $i+1 \leq q+1$). Outside the translates of $[p, q+1]$ the two elements are therefore given by the same rotations and agree. Inside, Lemma 1.1 gives

$$u_S : p \mapsto q+1, \quad j \mapsto j-1 \quad (p < j \leq q+1),$$

$$u_{S'} : p \mapsto i, \quad j \mapsto j-1 \quad (p < j \leq i), \quad i+1 \mapsto q+1, \quad j \mapsto j-1 \quad (i+1 < j \leq q+1).$$

These agree at every $j \in [p, q+1]$ except $j = p$ and $j = i+1$, which are distinct because $p \leq i < i+1$. The displayed values are read off directly, and $u_{S'}(p) = i \leq q < q+1 = u_S(p)$, $u_{S'}(i+1) = q+1 > i = u_S(i+1)$. $\square$

Lemma 2.2 (The two moved positions are incongruent). $p \not\equiv i+1 \pmod{n}$.

Proof. By (4), $i-p \leq q-p \leq n-2$, so $i+1-n \leq p-1 < p$; and $p \leq i < i+1$. Hence $i+1-n < p < i+1$, an open interval of length n whose only points congruent to $i+1$ modulo n are its endpoints. So $p \not\equiv i+1$. $\square$

Lemma 2.3 (Nothing below p climbs above $p-1$). *For every $x \leq p-1$ we have $u_S(x) \leq p-1$ and $u_{S'}(x) \leq p-1$.*

Proof. By (4), $p-1$ is a gap of S , hence also of $S' \subseteq S$. Let $[g+1, g']$ be the block of S (resp. of S') containing x , so $g < x \leq g'$ with g, g' consecutive gaps. If $g' > p-1$ then $g < x \leq p-1 < g'$, so the gap $p-1$ lies strictly between the consecutive gaps g and g' — impossible. Hence $g' \leq p-1$. By Lemma 1.1 the block is preserved, so $u(x) \in [g+1, g']$ and $u(x) \leq g' \leq p-1$. $\square$

3 The theorem

Theorem 3.1 (Deletion). *Let $n \geq 2$, let (S, T) be additive, and let $i \in S$. Then*

$$\ell(u_{S \setminus \{i\}} u_{T \setminus \{i+1\}}) = |S \setminus \{i\}| + |T \setminus \{i+1\}|,$$

i.e. $(S \setminus \{i\}, T \setminus \{i+1\})$ is additive.

Proof. Write $S' = S \setminus \{i\}$ and $T' = T \setminus \{i+1\}$, and fix an integer lift i of i with its S -run $[p, q]$ as in §2. Both S' and T' are proper subsets of $\mathbb{Z}/n$, and $\ell(u_{S'}) = |S'|$ by Lemma 1.1. By Lemma 1.2 applied with $w = u_{S'}$ and $X = T'$, it suffices to prove

$$u_{S'}(M'+1) > u_{S'}(j) \quad \text{for every run } [m', M'] \text{ of } T' \text{ and every } j \in [m', M']. \quad (5)$$

If $T' = \emptyset$ there is nothing to prove. Fix a run $[m', M']$ of T' ; by (1) we may translate it by any multiple of n , since (5) is invariant under that translation. By (2),

$$m' \geq M' - n + 2. \quad (6)$$

Two remarks are used repeatedly. First, every $j \in [m', M']$ lies in T' , so

$$j \not\equiv i+1 \pmod{n} \quad (j \in [m', M']), \quad (7)$$

because T' is the periodic lift of a set not containing the residue of $i+1$. Second, by Lemma 2.1 and (7),

$$u_{S'}(j) \leq u_S(j) \quad (j \in [m', M']), \quad (8)$$

with equality unless $j \equiv p$.

We distinguish cases according to the residue of $M' + 1$. By Lemma 2.2 the two conditions $M' + 1 \equiv p$ and $M' + 1 \equiv i + 1$ are mutually exclusive.

Case 2: $M' + 1 \equiv p \pmod{n}$. Translate the run so that $M' + 1 = p$, i.e. $M' = p - 1$. Then $u_{S'}(M' + 1) = u_{S'}(p) = i$ by Lemma 2.1. Every $j \in [m', M']$ satisfies $j \leq p - 1$, so $u_{S'}(j) \leq p - 1$ by Lemma 2.3. Since $p \leq i$ we get $u_{S'}(j) \leq p - 1 \leq i - 1 < i = u_{S'}(M' + 1)$, which is (5).

In the remaining cases $M' + 1 \not\equiv p$, so Lemma 2.1 gives

$$u_{S'}(M' + 1) \geq u_S(M' + 1). \quad (9)$$

Since $M' + 1 \notin T'$, either $M' + 1 \notin T$ or else $M' + 1 \in T$ and $M' + 1 \equiv i + 1$.

Case 1a: $M' + 1 \notin T$. Then M' is the top of the run $[a, b]$ of T containing $[m', M']$ (such a run exists because $T' \subseteq T$ and $[m', M']$ is an interval in T'), i.e. $b = M'$. Additivity of (S, T) and Lemma 1.2 applied with $w = u_S$ and $X = T$ give $u_S(b + 1) > u_S(j)$ for all $j \in [a, b]$, hence in particular

$$u_S(M' + 1) > u_S(j) \quad (j \in [m', M']).$$

Combining with (9) on the left and (8) on the right, $u_{S'}(M' + 1) \geq u_S(M' + 1) > u_S(j) \geq u_{S'}(j)$, which is (5).

Case 1b: $M' + 1 \in T$ and $M' + 1 \equiv i + 1 \pmod{n}$. Translate the run so that $M' + 1 = i + 1$, i.e. $M' = i$. Then $u_{S'}(M' + 1) = u_{S'}(i + 1) = q + 1$ by Lemma 2.1, and by (6) we have $m' \geq i - n + 2$. Let $j \in [m', i]$; we show $u_{S'}(j) < q + 1$, which is (5).

If $j \equiv p \pmod{n}$. Since $m' \geq i - n + 2$ and $p \leq i$, the points of $[m', i]$ congruent to p lie among $p - n, p, p + n$. Now $p + n > i$ because $p \geq i - n + 2$, and $p - n < m'$ because $p - n \geq m'$ would force $p \geq m' + n \geq i + 2$, contradicting $p \leq i$. So the only possibility is $j = p$ (and if $p < m'$ there is no such j at all, which is also fine); then $u_{S'}(p) = i \leq q < q + 1$.

If $j \not\equiv p \pmod{n}$. Then $u_{S'}(j) = u_S(j)$ by Lemma 2.1 together with (7). If $p < j \leq i$ then $j - 1 \in [p, i - 1] \subseteq S$, so $u_S(j) = j - 1 \leq i - 1 < q + 1$ by Lemma 1.1. If $j < p$ then $u_S(j) \leq p - 1 < q + 1$ by Lemma 2.3. (The case $j > i$ does not occur, as $j \leq M' = i$; and $j = p$ was treated above.)

The three cases are exhaustive, so (5) holds for every run of T' , and Lemma 1.2 gives $\ell(u_{S'}u_{T'}) = \ell(u_{S'}) + |T'| = |S'| + |T'|$. $\square$

Corollary 3.2 (Q232, bracketed-pair deletion). *Let $n \geq 3$, let (S, T) be additive, and suppose $i \in S$ and $i + 1 \in T$. Then*

$$\ell(u_{S \setminus \{i\}} u_{T \setminus \{i+1\}}) = |S| + |T| - 2.$$

Proof. Theorem 3.1 with $|S \setminus \{i\}| = |S| - 1$ (as $i \in S$) and $|T \setminus \{i + 1\}| = |T| - 1$ (as $i + 1 \in T$). $\square$

4 The bracket hypothesis is not load-bearing

The brief for this session asserted that the hypothesis $i + 1 \in T$ must be consumed by any correct proof, on the ground that the *unbracketed analogue* — additivity is hereditary under $S' \subseteq S$, $T' \subseteq T$ — is false, with smallest witness $n = 3$, $S = \{0\}$, $T = \{0, 1\}$, $S' = T' = \{0\}$. That inference does not go through, and it is worth saying exactly why, because the conclusion drawn from it (“any proof that does not use the bracket is wrong”) would have rejected the proof above.

The false analogue and Theorem 3.1 differ in *which* elements are deleted, not in how many. In the witness $S = \{0\}$, $T = \{0, 1\}$, $S' = T' = \{0\}$, nothing is deleted from S at all; the deletion is $T \mapsto T \setminus \{1\}$ with S held fixed. A second witness from the same census, $n = 3$, $S = \{0, 1\}$, $T = \{1\}$, $S' = \{1\}$, $T' = \{1\}$, is the mirror failure: here $i = 0 \in S$ is deleted from S and $i + 1 = 1 \in T$ is *not* deleted from T , and $(\{1\}, \{1\})$ is indeed not additive ($u_{\{1\}}u_{\{1\}} = s_1s_1 = \text{id}$, of length $0 \neq 2$). What these show is that the two deletions must be *coupled*: removing i from S obliges one to remove $i + 1$ from T if it is there. They say nothing about whether $i + 1$ must be in T to begin with.

Theorem 3.1 is exactly this coupled statement, and the coupling is what the proof consumes, at (7): deleting i from S *raises* the value of u_S at the single position $i + 1$ (Lemma 2.1), which is the only way the criterion of Lemma 1.2 could be broken on the right-hand side, and that position is unavailable as a j precisely because $i + 1$ was deleted from T . So the bracket appears in the proof as the *form of the conclusion*, not as a hypothesis: it is the reason T' is $T \setminus \{i + 1\}$ rather than some other subset. When $i + 1 \notin T$ the deletion is empty, $T' = T$, and the same argument shows $(S \setminus \{i\}, T)$ is additive. A census confirms this is not vacuous: over $3 \leq n \leq 8$ there are 60 082 additive triples (S, T, i) with $i \in S$ and $i + 1 \notin T$, and $(S \setminus \{i\}, T)$ is additive in all of them (§5).

Remark 4.1. The structural reason the coupled statement is so much better behaved than the general hereditary one is visible in Lemma 2.1. Deleting a single element from S is a minimal perturbation of u_S : it moves exactly two values, and it moves them in opposite directions. The criterion of Lemma 1.2 is monotone — it wants the value at $M' + 1$ large and the values on $[m', M']$ small — so a *lowered* value is harmless at j and a *raised* value is harmless at $M' + 1$. Of the four position/direction combinations, three are therefore automatically benign, and the fourth (the raised value appearing as some j) is killed by the deletion from T . What remains is the single genuinely new case, Case 2, where the lowered value appears at $M' + 1$; and there Lemma 2.3 applies because the position in question is the bottom of a block, so everything below it is trapped below it.

5 Computational verification

All code is in `proofs/code-q232/` and `reviews/code-2026-09-22-c2/` of the commit named on the title page. Affine permutations are held in window notation and lengths are computed by Shi's inversion formula, not by the block model, so Lemma 1.1 is tested rather than assumed.

The statement. `g1_lemma.py` re-run on 23 September 2026: over all additive pairs with $3 \leq n \leq 8$ and all i with $i \in S$, $i + 1 \in T$, the reduced pair is additive in 82 044 of 82 044 instances. `g1_lemma2.py` re-run the same day: the hereditary analogue fails in 412 578 of 1 822 537 instances for $3 \leq n \leq 7$. Both counts reproduce those recorded in [2, §Gaps, (G1)] exactly.

The strengthening. `code-q232/stronger.py`: over $3 \leq n \leq 8$, all additive (S, T) and all $i \in S$ with $i + 1 \notin T$, the pair $(S \setminus \{i\}, T)$ is additive in 60 082 of 60 082 instances, 0 failures.

The mechanism, not only the conclusion. `code-q232/mechanism.py` checks the individual structural assertions of the proof rather than its conclusion, over all additive (S, T) with $2 \leq n \leq 8$ and all $i \in S$ (142 130 instances):

- Lemma 2.1: $u_{S'}$ and u_S differ at x if and only if $x \equiv p$ or $x \equiv i + 1$, and the four displayed values are as stated — 142 130 of 142 130, no exception;
- Lemma 2.2: $p \not\equiv i + 1$ — 142 130 of 142 130;
- Lemma 2.3: $u_S(x), u_{S'}(x) \leq p - 1$ for all $x \leq p - 1$ — 142 130 of 142 130;
- the case split is exhaustive and no branch is vacuous: every run of T' falls in exactly one of Cases 1a, 1b, 2, with 151 052, 38 192 and 64 920 occurrences respectively, and the assertion “Case 1b $\Rightarrow M' + 1 \equiv i + 1$ ” never failed;
- the criterion (5) holds on every run in every case (0 failures), and the case-specific reasons hold too: in Case 2, $u_{S'}(j) \leq p - 1$ throughout the run; in Case 1a, $u_{S'}(j) \leq u_S(j)$ on the run and $u_{S'}(M' + 1) \geq u_S(M' + 1)$;
- Lemma 1.2 agrees with Shi’s formula on all 142 130 reduced pairs and on every original pair, so the criterion used in the proof is the same predicate that the censuses measure.

6 Consequence: gap (G1) closes

Theorem and section numbers of [1, 2] quoted below are those of the build at the commit named on the title page; the internal labels, which are stable, are `lem:blocks` = Lem. 2.2 and `lem:add` = Lem. 2.3 and `thm:exchange` = Thm. 4.1 and `lem:apply` = Lem. 4.2 in [1], and `def:E` = Def. 1.1 and `lem:step1` = Lem. 3.1 and `thm:contain` = Thm. 3.2 and `sec:emptyX` = §5 in [2].

Gap (G1) of [2] records [2, §5] (the case $\mathcal{X}(S, T) = \emptyset$) as computational rather than proved, and names the obstruction precisely: the proof of [2, Thm. 3.2(iii)] invokes [2, Lem. 3.1] (Step 1 of [1, Lem. 4.2]), which requires its pair to be additive, whereas the $\mathcal{X} = \emptyset$ construction of Morse–Schilling removes a bracketed pair $(b - 1, b)$ with $b - 1 \in S$ and $b \in T$ and proceeds with the *reduced* pair $(S \setminus \{b - 1\}, T \setminus \{b\})$.

Corollary 3.2 with $i = b - 1$ says exactly that the reduced pair is additive. We now show that this is enough: the two computational bullets of [2, §5] become theorems.

Definition 6.1 (the construction). Let (S, T) be additive with $S \cup T = \mathbb{Z}/n$. A *Case-(3) index* is a $b \in \mathbb{Z}/n$ for which there is a $t \geq 1$ with

$$[b - t, b - 1] \subseteq S, \quad b \notin S, \quad [b - t + 1, b] \subseteq T, \quad b - t \notin T.$$

Put $S_o = S \setminus \{b - 1\}$ and $T_o = T \setminus \{b\}$. If $\tilde{e}_1^{(b)}(u_{S_o} u_{T_o}) \neq \mathbf{0}$, equivalently $L_1^{(b)} \neq \emptyset$ for the pairing of (S_o, T_o) at cut b , let $c = \min_{\prec_b} L_1^{(b)}$ and let m be the bottom of the run of S containing c (*reading A*) or of the run of S_o containing c (*reading B*). The move produced is $(S, T) \mapsto (S \setminus \{c\}, T \cup \{m\})$.

Theorem 6.2 ((G1’): the transport). *In the situation of Definition 6.1, with $L_1^{(b)} \neq \emptyset$:*

- (1) *readings A and B give the same m ;*
- (2) $|T| \leq n - 2$;
- (3) *the run $[m, M]$ of S containing c is usable, $c = e_{[m, M]}$, and hence $c \in \mathcal{E}(S, T)$ and $(S \setminus \{c\}, T \cup \{m\}) \in \mathcal{E}^{\text{mv}}(S, T)$;*
- (4) $u_{S \setminus \{c\}} u_{T \cup \{m\}} = u_S u_T$, *and this factorisation is length-additive.*

Proof. Step 0 (Theorem 3.1 sets the machine running). $S_o \subseteq S$ and $T_o \subseteq T$ are proper. Since $b - 1 \in S$ and $b \in T$, Corollary 3.2 with $i = b - 1$ gives that (S_o, T_o) is additive. Since $b \notin S$ we have $b \notin S_o$, and $b \notin T_o$ by construction, so $b \in \mathcal{X}(S_o, T_o) = \mathbb{Z}/n \setminus (S_o \cup T_o)$. Therefore [2, Thm. 3.2] applies to the additive pair (S_o, T_o) at $x = b$, and with c as above and $[m_o, M_o]$ the run of S_o containing c it yields

$$(i) \ c + 1 \notin T_o, \quad (ii) \ [m_o + 1, c] \subseteq T_o, \quad (iii) \ m_o \notin T_o. \quad (10)$$

This is the only place the deletion theorem is used, and it is the whole of its contribution.

Step 1 (the readings agree). Because $b \notin S$ and $b - 1 \in S$, the element $b - 1$ is the *top* of its run in S , say $[m_1, b - 1]$. Deleting the top of a run cannot split it: the runs of S_o are exactly the runs of S with $[m_1, b - 1]$ replaced by $[m_1, b - 2]$ (and deleted outright if $m_1 = b - 1$). In particular every run of S_o has the same bottom as the run of S containing it. Since $c \in S_o$, the run of S containing c is the run of S_o containing c , possibly extended at the top from $b - 2$ to $b - 1$; so $m = m_o$ under either reading, and $M \in \{M_o, b - 1\}$. This proves (1), and also identifies m with the m_o of (10).

Step 2 ($|T| \leq n - 2$). Suppose not. Since $b - t \notin T$, T is proper, so $|T| = n - 1$ and T has a single gap class g . As $b \in T$ we have $g \neq b$, so $T_o = T \setminus \{b\}$ has exactly the two gap classes g and b . By (10)(iii), $m_o \notin T_o$, so $m_o \equiv g$ or $m_o \equiv b$; the latter is impossible because $m_o \in S_o \subseteq S$ while $b \notin S$. Hence $m_o \equiv g$. By (10)(i), $c + 1 \notin T_o$, so $c + 1 \equiv g$ or $c + 1 \equiv b$; the latter would give $c \equiv b - 1$, impossible because $c \in S_o = S \setminus \{b - 1\}$. Hence $c + 1 \equiv g \equiv m_o$. But c lies in the run $[m_o, M_o]$, whose length is at most $n - 1$ by (2), so $c + 1 \in [m_o + 1, m_o + n - 1] = n - 1$ consecutive integers, none congruent to m_o modulo n . Contradiction.

Step 3 (transport). Let $[m, M]$ be the run of S containing c ; $m = m_o$ by Step 1.

- $m \notin T$: by (10)(iii) $m \notin T \setminus \{b\}$, so $m \notin T$ unless $m \equiv b$; but $m \in S$ and $b \notin S$.
- $c + 1 \notin T$: by (10)(i) $c + 1 \notin T \setminus \{b\}$, so $c + 1 \notin T$ unless $c + 1 \equiv b$; but that gives $c \equiv b - 1 \notin S_o$.
- every $j \in [m, c - 1]$ has $j + 1 \in T$: immediate from (10)(ii), since $[m_o + 1, c] \subseteq T_o \subseteq T$.

The first and second points make $[m, M]$ usable ($m \notin T$, and $c \in [m, M]$ has $c + 1 \notin T$); the third and second make c the *first* $j \in [m, M]$ with $j + 1 \notin T$. So $c = e_{[m, M]}$ in the sense of [2, Def. 1.1], which is (3). Note that (3) uses only the combinatorial definition of $\mathcal{E}^{\text{mv}}$ and not Step 2.

Step 4 (the move is the factorisation). By Step 2, $|T| \leq n - 2$; by Step 3 the run $[m, M]$ of S is usable with first index c . So [1, Lem. 4.3] applies to (S, T) and the run $[m, M]$, giving $u_{S \setminus \{c\}} u_{T \cup \{m\}} = u_S u_T$ with the factorisation length-additive. This is (4). $\square$

Corollary 6.3. *The two assertions of [2, §5] recorded there as computational — that every move produced by the $\mathcal{X} = \emptyset$ construction is a genuine length-additive factorisation of $u_S u_T$, and that every such move lies in $\mathcal{E}^{\text{mv}}(S, T)$ — hold for every n , every additive (S, T) with $S \cup T = \mathbb{Z}/n$ and every Case-(3) index b . Gap (G1) is closed, and the reading ambiguity of gap (G2) between readings A and B is immaterial, by Theorem 6.2(1).*

Remark 6.4 (what is still not proved). Corollary 6.3 does not touch the *exegetical* half of (G2): that Definition 6.1 is what [4, §two factor case] intends by “ t maximal” and “proceed with $x = b$ ” remains my reading of their prose, and no theorem about $\mathbb{Z}/n$ can settle it. What Theorem 6.2(1) does settle is that the one ambiguity I was able to make precise — whether the offset is computed against $\text{content}(u)$ or against the reduced content — has no effect on the answer, which [2, §Gaps, (G2)] could only record as evidence. Nor does it touch the strictness

claim of [2, §5] (equality fails in 7 pairs, all at $n = 7$); that is an existence statement, certified by its witnesses.

Remark 6.5 (why Step 2 is not cosmetic). The bound $|T| \leq n-2$ is the hypothesis of [1, Lem. 4.3] and it genuinely fails in the ambient regime: over $3 \leq n \leq 8$ there are 6786 pairs (additive, $\mathcal{X} = \emptyset$) with a Case-(3) index, and in 894 of them $|T| = n-1$. What Step 2 shows is that those 894 are precisely instances where the construction produces nothing — and the computation confirms it: $L_1^{(b)} = \emptyset$ in all 894, and $\mathcal{E}(S, T) = \emptyset$ in all 894. Had I taken $|T| \leq n-2$ for granted, the proof would have been wrong on a set of positive density without ever producing a false move.

References

- [1] Clio, *An exchange move for cyclically decreasing factorisations, and the reduction of WZZ Problem 5.1 to a single hypothesis*, 21 September 2026, [clio-vega/proofs@22d1b22](#).
- [2] Clio, *Every Morse–Schilling crystal edge is an exchange move, and the exchange move is strictly more general*, 22 September 2026, [clio-vega/proofs@eb9d081](#).
- [3] T. Lam, *Affine Stanley symmetric functions*, Amer. J. Math. 128 (2006), 1553–1586.
- [4] J. Morse and A. Schilling, *Crystal approach to affine Schubert calculus*, [arXiv:1408.0320](#).